

Case study Data Analyst

Please provide your code and any other helpful material for review at least a day before the interview.

Adevinta is all about bringing buyers and sellers together to make a great deal. Sellers place ads, and buyers look for these ads by browsing or searching on the site. When a buyer finds an ad they find interesting in the result list and clicks it, we call this a 'View Item page' – a VIP view. When a buyer proceeds to contact a seller to get more info or strike a deal, we call this a lead. They can do so by calling (Phone click), asking a question over email (ASQ: ask seller Question), clicking out to a seller's website (URL_CLICK) or placing a bid.

- 1) Basic querying: Given the following tables, write a SQL script to return the total number and percentage of ads per category which receive leads within a period of 24 hours in May 2017

Ad_table		
Field	Type	Example value
seller_id	int	1673288
ad_id	int	1174460500
ad_title	string	Volkswagen Golf 1.6 Variant 77KW 2003 Grijs NAP
ad_start_date	date	4/1/2017
ad_start_time	time	10:25:11
ad_end_date	date	4/29/2017
ad_end_time		17:54:21
category	string	Auto's
subcategory	string	Volkswagen
Event_table		
Field	Type	Example value(s)
clsfd_ad_id	int	1174460500
clsfd_seller_id	int	1673288
buyer_id	int	548936
date	date	4/2/2017
timestamp	time	16:31:13
eventtype	string	VIP_VIEW;PHONE_CLICK;EMAIL;BID; URL_CLICK
n_events	int	1

Attached is a dataset containing information on Car ads in our Dutch market (Marktplaats)

fieldname	explanation
src_ad_id	id of the ad
telclicks	number of phone clicks
bids	number of bids
carrosserie	car type
photo_cnt	number of photos
aantaldeuren	number of doors
n_asq	number of emails sent to seller
bouwjaar	year of build of the car
emissie	emission
energielabel	energy label
brand	brand
ad_start_dt	ad start date
vermogen	power
webclicks	number of clicks to seller website
model	model
aantalstoelen	number of chairs
price	price
test group	car was in testgroup a, b, or in no test

2) On testing: Marktplaats has decided on implementing a product feature that we think will drive the number of Car ads with leads: How would you ideally set up the test whether this initiative has improved this metric?

- Why do you choose this method?
- What are the risks / challenges?
- Use the dataset attached to examine the test groups:
 - o What is the result of your analysis?
 - o What insights did you gain?

3) On insights and use cases:

- What first impressions do you get from this data?
- Name three use cases for this dataset (assume you might also combine it with other public or Adevinta datasets)